

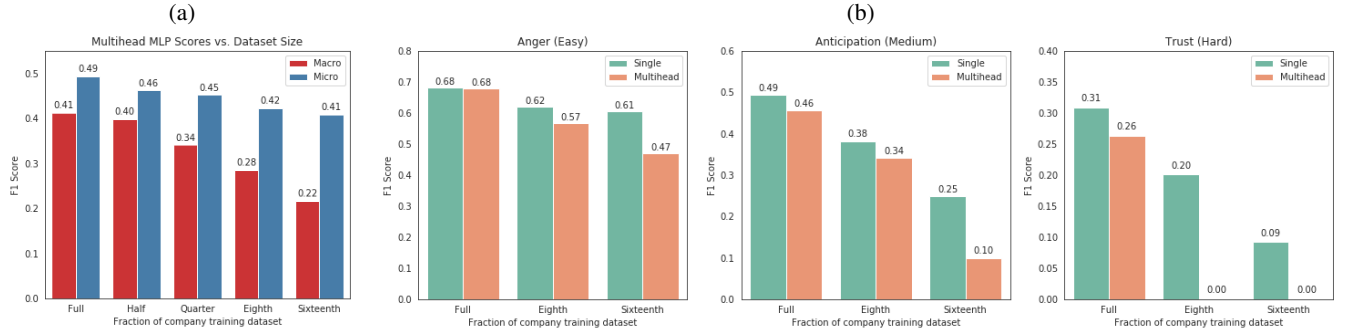


Figure 2: a) Comparison of macro and micro averages of F1 scores across categories on the company tweets dataset. b) F1 Scores for different categories on different dataset sizes for single head vs. multihead decoder.

the dataset to contain difficult and subtle examples of sentiments, not just those examples where everyone agrees. The raters also have a tendency to under-label categories, especially when presented multiple options.

Following a similar process, we required 2 out of 5 raters for a positive label, and in the case of binary sentiment labels (Positive, Neutral, Negative), we rounded toward polarized sentiment and away from Neutral labels in the case of a 2/3 split. Applying the SemEval-trained Transformer directly to our company tweets dataset gets reasonably good results (0.338 macro average), also validating that our labeling technique is similar to that of SemEval.

Looking at rater agreement by dataset (Fig. 3), we see that Plutchik category labels contain large rater disagreement, even among vetted raters who passed the golden set test. Furthermore, datasets with more emotions (the SemEval dataset and our active learning sampled company tweets) contain higher Plutchik disagreement than random company tweets. This is likely because raters tend to apply the "No Emotion" label when they are not sure about a category. As Table 2 shows, the SemEval and active company tweets datasets contain fewer no-emotion tweets than other datasets.

It would be interesting to analyze rater disagreement by category, how much this effects classifier convergence, whether getting 7+ ratings per tweet helps classifier convergence, and also whether this work could benefit from estimating rater quality via agreement with the crowd, as proposed in (?). However this analysis is not straightforward, as the truth data is itself collected through human labeling.

Alongside classifier convergence by dataset size (Fig. 2b), we think that this could be an interesting area a future research.

Difficult tweets and challenging contexts. There is not sufficient space for a thorough analysis, but we wanted to suggest why general purpose APIs may not work well on our company tweets dataset. Table 1 samples the largest binary sentiment disagreements between human raters and the Watson API. For simplicity, we restrict examples to video game tweets, which comprise 19.1% of our test set. As we can see, all of these examples appear to ascribe negative emotion to generally negative terms which, in a video game context, do

not indicate negative sentiment.

Our purpose is not to castigate the Watson or the GCL APIs. Rather, we propose that it may not be possible to provide context-independent emotion classification scores that work well across text contexts.

It may work better in practice, on some tasks, to train a large unsupervised model and to use a small amount of labeled data to finetune on the context present in the specific dataset. We would like to quantify this further in future work.

Recent work (?) shows that training an RNN with multiple softmax outputs leads to a much improved BPC on language modeling, especially for diverse datasets and models with large vocabularies. This is because the multiple softmaxes are able to capture a larger number of distinct contexts in the text than a single output.

Perhaps our Transformer also captures the features relevant to a large number of distinct contexts, and the finetuning is able to select the most significant of these features, while ignoring those features that – while adding value in general – are not appropriate in a video game setting.

Conclusion

In this work we demonstrate that unsupervised pretraining and finetuning provides a flexible framework that is effective for difficult text classification tasks. We noticed that the finetuning was especially effective with the Transformer network, when transferring to downstream tasks with noisy labels and specialized context.

We think that this framework makes it easy to customize a text classification model on niche tasks. Unsupervised language modeling can be done on general text datasets, and requires no labels. Meanwhile downstream task transfer works well enough, even on small amounts of domain-specific labelled data, to be accessible to most academics and small organization.

It would be great to see this approach applied to a variety of practical text classification problems, much as (?) and (?) have applied language modeling and transfer to a variety of academic text understanding problems on the GLUE Benchmark.

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